



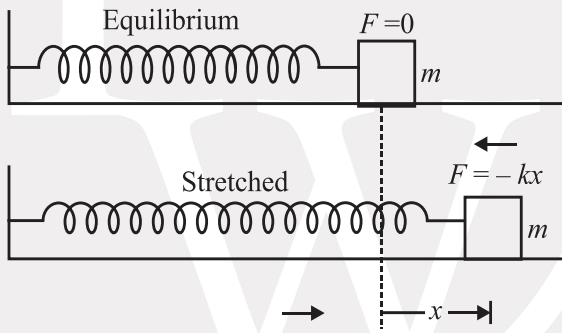
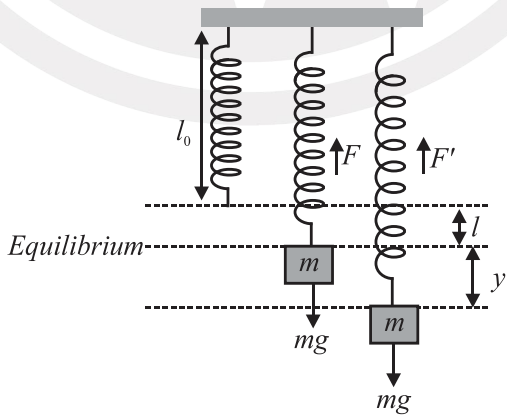
CHAPTER

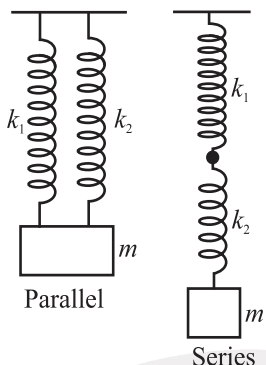
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Oscillations

Sr. No.	Concept	Formulas	Other information
1.	Basic terms related to periodic function	Displacement, $x = A \sin \omega t$ Frequency (ν) = $\frac{1}{\text{Time period } (T)}$ Angular frequency (ω) = $\frac{2\pi}{\text{Time period } (T)}$ $= 2\pi \times \text{Frequency } (\nu)$ A combination of sine & cosine function is also a periodic function $x = a \sin \omega t + b \cos \omega t$ Take $a = R \cos \phi$ and $b = R \sin \phi$, Then $x = R \sin (\omega t + \phi)$ $R = \sqrt{a^2 + b^2}$, $\tan \phi = \frac{b}{a}$, $\phi = \tan^{-1} \left(\frac{b}{a} \right)$	A = amplitude of oscillations
2.	Displacement, velocity, acceleration and time period of SHM	Displacement, $x = A \sin (\omega t + \phi_0)$ Velocity, $v = \frac{dx}{dt} = \omega A \cos (\omega t + \phi_0)$ $= \omega \sqrt{A^2 - x^2}$ Maximum velocity, $v_{\max} = \omega A$. Acceleration, $a = \frac{dv}{dt} = -\omega^2 A \sin (\omega t + \phi_0) = -\omega^2 x$ Maximum acceleration, $a_{\max} = \omega^2 A$ Time period, $T = 2\pi \sqrt{\frac{\text{Displacement}}{\text{Acceleration}}} = 2\pi \sqrt{\frac{x}{a}}$	A = amplitude of SHM ω = angular frequency ϕ_0 = initial phase of particle in SHM. k = force constant t = time a = acceleration

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3.	Force law for SHM and expression for time period	Restoring force, $F = -kx = -m\omega^2 x$ And $\omega^2 = \frac{k}{m}$ Angular frequency, $\omega = 2\pi\nu = \frac{2\pi}{T}$ Time period, $T = 2\pi \sqrt{\frac{\text{Inertia factor}}{\text{Spring factor}}} = 2\pi \sqrt{\frac{m}{k}}$	m = mass of the body ν = frequency of oscillation T = time period
4.	Kinetic Energy of SHM	(i) In terms of displacement: $KE = \frac{1}{2}m\omega^2(A^2 - x^2)$ As, $\omega = \sqrt{\frac{k}{m}} \Rightarrow m\omega^2 = k$ $\therefore KE = \frac{1}{2}k(A^2 - x^2)$ Kinetic energy is maximum at mean position, i.e. at $x = 0$ $KE_{\max} = \frac{1}{2}m\omega^2 A^2$ Kinetic energy is minimum (zero) at extreme position, i.e. at $x = \pm A$, $KE_{\min} = \text{zero}$ (ii) In terms of time: $K = \frac{1}{2}m\omega^2 A^2 \cos^2 \omega t$	m = mass of the body A = amplitude of SHM x = displacement from mean position $(x = A \sin \omega t)$
5.	Potential energy of SHM	In terms of displacement, $\text{Potential energy, } U = \frac{1}{2}m\omega^2 x^2 = \frac{1}{2}kx^2$ Potential energy is maximum at extreme position, i.e. at $x = \pm A$ $U_{\max} = \frac{1}{2}m\omega^2 A^2$ Potential energy is minimum (zero) at mean position, i.e. at $x = 0$, $U_{\min} = 0$ In terms of time; $U = \frac{1}{2}m\omega^2 A^2 \sin^2 \omega t$	U = potential energy m = mass of the body ω = angular frequency of SHM A = amplitude of SHM $x = A \sin \omega t$
6.	Total energy of SHM	For a simple harmonic motion; $E = K + U = \frac{1}{2}m\omega^2(A^2 - x^2) + \frac{1}{2}m\omega^2 x^2$ $E = \frac{1}{2}m\omega^2 A^2$ or $E = \frac{1}{2}kA^2$	E = total energy of SHM k = force constant K = kinetic energy U = potential energy

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7.	Oscillations of a simple pendulum	Time period, $T = 2\pi\sqrt{\frac{l}{g}}$ Frequency, $\nu = \frac{1}{2\pi}\sqrt{\frac{g}{l}}$	l = length of the pendulum g = gravitational acceleration
8.	Practical applications of SHM	For a liquid of density ρ contained in a U-tube upto height h , $T = 2\pi\sqrt{\frac{h}{g}}$ For a body dropped in a tunnel along the diameter of the earth, $T = 2\pi\sqrt{\frac{R}{g}}$ For a cylinder of density ρ floating with length h submerged in a liquid of density σ , $T = 2\pi\sqrt{\frac{\rho h}{\sigma g}}$	R = radius of earth T = time period of oscillation g = gravitational acceleration
9.	Oscillations of a loaded spring	Vibrations of horizontal string;  Vibration of a vertical spring:  In both the cases mentioned above,	k_1, k_2 = spring constants of two springs. k = equivalent spring constant T = time period of oscillations x = extension produced in each spring. F = restoring force

Sr. No.	Concept	Formulas	Other information
		Period of oscillation $T = 2\pi\sqrt{\frac{m}{k}}$  Parallel Series For two springs of springs factors k_1 and k_2 connected in parallel, effective spring factor, $k = k_1 + k_2$ $\therefore T = 2\pi\sqrt{\frac{m}{k_1 + k_2}}$ For two springs connected in series, effective spring factor k is given by $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} \quad \text{or} \quad k = \frac{k_1 k_2}{k_1 + k_2}$ $\therefore T = 2\pi\sqrt{\frac{m(k_1 + k_2)}{k_1 k_2}}$ When a spring is cut into n equal pieces, spring factor of each part becomes nk. $\therefore T = 2\pi\sqrt{\frac{m}{nk}}$	